Roseline Bandela

bandelaroseline@gmail.com | +1 337.244.3257 | linkedin.com/in/roseline-bandela | Lake Charles, LA

EDUCATION

ALFRED M. BARBE HIGH SCHOOL – *TOPS Diploma* Graduation: May 2027

GPA: 4.23 | Total Credits as of 11th Grade: 30.5

- AP Coursework (Senior Year): AP Chemistry with Lab, AP Environmental Science, AP Literature & Composition, AP US History
- Self-Studied AP Exams (Senior Year): AP Calculus BC, AP Biology, AP Physics C: Mechanics, AP Physics C: Electricity & Magnetism
- Prior Coursework: AP Calculus AB, AP Pre-Calculus, AP Computer Science A, AP Statistics, AP Physics 1, AP Research, AP Seminar, AP US Government, AP Human Geography

MCNEESE STATE UNIVERSITY – *Dual Enrollment* Current

- Coursework: BIOL 101/101L (Fall Semester); MATH 190 and ENGR 111 (Spring Semester)

LSU STEM PATHWAYS – *Gold Seal Pre-Engineering Pathway* Expected 2027

- Completed Intro to Robotics, Engineering Design & Development, and Principles of Engineering; completing Intro to Engineering to earn the LSU Gold Seal

EXPERIENCE

WAVEGATE CORPORATION – *Biomedical Engineering Intern* Lake Charles, LA · Nov 2025 – Present

- Program embedded firmware for medical device prototypes using Code Composer Studio and the MSP430 microcontroller
- Configure and validate the ADPD4100 optical analog front-end for biomedical signal acquisition, analyzing hexadecimal sensor outputs to verify function and improve firmware
- Assist in hardware integration, debugging, embedded testing, and communication between biomedical sensors and microcontrollers
- Collaborate with engineers developing the Ellipse™ Spinal Cord Stimulator Platform

STANFORD IGEM BIOENGINEERING RESEARCH PROGRAM – *Teaching Assistant* July 2026

- Mentored four student bioengineering research teams in developing research proposals, refining project objectives, and translating scientific ideas into feasible research plans.
- Provided technical guidance on literature reviews, experimental design, and bioengineering methodologies, supporting students in strengthening the scientific rationale behind their projects.
- Guided students in communicating complex scientific concepts, structuring research proposals, and developing projects grounded in scientific evidence.

STANFORD IGEM BIOENGINEERING RESEARCH PROGRAM – *Research Team Leader* June - Aug 2025

- Led an international team of seven researchers investigating AI-based retinoblastoma screening using retinal fundus imaging
- Coordinated research planning, literature review, model development, and scientific communication
- Authored a full research paper with my team aiming to solve retinoblastoma in children under the age of 5 years old.

THINKNEURO, LLC – *Summer Research Intern* May - Aug 2025

- Investigated reinforcement learning approaches for adaptive neuroprosthetic systems and brain-computer interfaces
- Compared reinforcement learning algorithms with supervised learning methods for dynamic biomedical applications

Scanned with CamScanner

RESEARCH

AI-BASED EARLY DETECTION OF DIABETIC RETINOPATHY USING DEEP LEARNING – *Independent Research* 2025

- Designed a CNN-based pipeline for automated diabetic retinopathy screening in Python and TensorFlow
- Processed over 3,000 retinal fundus images from the APTOS 2019 dataset
- Implemented preprocessing, augmentation, model training, Grad-CAM visualization, ROC analysis, and evaluation metrics
- Authored a full research paper and presentation focused on affordable AI-enabled screening systems for underserved communities – in the middle of publishing the paper.
- Qualified for the Louisiana State Science & Engineering Fair

NEUROVESTIBULAR-OCULAR-SENSORIMOTOR DIGITAL TWIN FOR AUTONOMOUS ASTRONAUT TRAINING
Independent Research 2026

- Awarded a $7,000 research grant to fund development of the project
- Developed a computational digital-twin model of the neurovestibular, ocular, and sensorimotor systems to simulate astronaut spatial disorientation linked to Spaceflight-Associated Neuro-ocular Syndrome (SANS)
- Designed autonomous training modules that adapt countermeasure protocols in real time based on simulated sensorimotor feedback
- Applied physiological modeling and machine learning to generate individualized pre-flight training and rehabilitation recommendations
- Incorporating the PhysioNet Human Balance Evaluation dataset (Santos & Duarte) to ground the model in real data
- Working with a faculty sponsor to complete Phase 1 for science fair and support future grant applications

PROJECTS

LIFELINE – DISASTER CONTINUITY OF CARE APP
Congressional App – Rep. Clay Higgins (LA-03)

Founder and App Developer 2026

- Developed a healthcare-access application designed to help Louisiana residents maintain continuity of care before, during, and after hurricanes and other natural disasters.
- Addressed challenges faced by Southwest Louisiana communities, where recurring hurricanes can disrupt access to healthcare facilities, pharmacies, essential medical information, and ongoing treatment.
- Designed a community-centered digital solution to help individuals navigate healthcare disruptions, locate essential resources, and better prepare for medical needs during emergencies.
- Applied principles of biomedical engineering, human-centered design, and technology development to address a regional healthcare challenge with broader implications for disaster-prone communities worldwide.

NASA SPACE APPS CHALLENGE – *Team Participant* Virtual · 2025

- Built an Arduino-based astronaut biomedical monitoring system for real-time physiological tracking during long-duration missions

GESTURE-CONTROLLED MEDICAL IMAGING INTERFACE – *Independent Project* 2026

- Developed a touchless, gesture-based control system enabling clinicians to navigate medical imaging displays (e.g., ultrasound/MRI views) without breaking the sterile field
- Implemented real-time hand-gesture recognition using computer vision (OpenCV) to translate gestures into imaging-viewer commands such as zoom, pan, and slice navigation
- Prototyped the interface on an embedded platform to enable low-cost, low-latency touchless interaction in clinical settings

ZEOLITE WATER FILTRATION RESEARCH – *Independent Research* 2024

- Designed an environmentally sustainable ammonia-removal filtration system
- Earned 1st Place Regional Science Fair and 3rd Place Louisiana State Science Fair

HACK CLUB ONBOARD – PRINTED CIRCUIT BOARD DESIGN – *Independent Project* 2025

- Designed and fabricated a custom PCB through Hack Club's OnBoard hardware program, gaining hands-on experience in circuit design and fabrication

Scanned with CamScanner

LEADERSHIP, SERVICE & ATHLETICS

BARBE HIGH SCHOOL SCIENCE OLYMPIAD – *Founder, Team Captain & President* Present

- Founded a Science Olympiad team to establish a competitive, interdisciplinary STEM program spanning biology, chemistry, physics, earth science, and engineering.

SOUTHWEST LOUISIANA BIOELECTRONICS LABORATORY (SLaBEL) – *Founding Director* 2026 – Present

- Founding a selective student research organization at McNeese focused on bioelectronics, neural engineering, and brain-computer interfaces
- Secured support from McNeese's Dean of Engineering & Sciences, including a connection to a faculty advisor
- Designed a two-semester curriculum progressing from foundational skills to track-based research, and a constitution with a formal succession plan

VEX ROBOTICS TEAM KRAKEN 43146K – *Lead Programmer & Marketing Lead • Mechanical Assist* 2024 – Present

- Developed autonomous and driver-control software in C++ as lead programmer
- Helped qualify for the VEX Robotics World Championship twice; ranked Top 30 worldwide
- Contributed to Engineering Notebook development, marketing strategy, and robot optimization
- Designed the team's branding and built its website

MCNEESE STATE UNIVERSITY IGEM TEAM – *Founder & Team Lead* 2025

- Organize software, wet lab, and dry lab teams
- Coordinate project timelines, outreach, sponsorship, and research direction

NY JUNIOR ACADEMY OF SCIENCES – *Project Manager* New York, NY

- Lead a six-member international team designing a solar-powered prosthetic device for underserved communities

TRINITY BAPTIST CHURCH – *Volunteer*

- Volunteer at Trinity Baptist Church gaining around 100 hours of service with Vacation Bible School, WinShape Camp, and Sunday volunteering.
- Arts & Crafts main lead for After-VBS Camp
- Assisted with infant care

BARBE HIGH SCHOOL WRESTLING – *Varsity Wrestler* 2025-Present

- Varsity Women's Wrestling – competing in the 100 lb category at 90 lbs

HONORS & AWARDS

- Yale Science & Engineering Association – Outstanding STEM Exhibit
- Regeneron Biomedical Science Award
- Society for In Vitro Biology - Outstanding Achievement Award
- RICOH USA Award
- Stockholm Junior Water Prize Award
- Environmental Sustainability Award
- AP Scholar Award (AP Seminar + AP Research)

SPECIFIC TO VEX V5 ROBOTICS:

VEX V5 High Stakes (2024-25)

- Energy Award, VEX Robotics World Championship (2025)
- Excellence Award, Louisiana State Championship (2025) – World Championship Qualified
- Excellence Award, TORS Tournament (2025)
- Tournament Champion, Leagues Finals (2024)
- Design Award, Leagues Finals (2024)

VEX V5 Push Back (2025-26)

- Tournament Finalist, Louisiana State Championship (2026)
- Design Award, Louisiana State Championship (2026) – World Championship Qualified
- Energy Award, Rodeo DFW Signature Event (2026)
- Tournament Champion, Leagues Finals (2025)
- Robot Skills Champion, Leagues Finals (2025)
- Judges Award, Leagues Finals (2025)

Scanned with CamScanner

- Judges Award, Space City (2025)
- Design Award, Holly Jolly (2025)

SKILLS & INTERESTS

Programming: Python, C++, JavaScript, React, R, MATLAB | **Machine Learning:** TensorFlow, PyTorch, OpenCV, CNNs, Grad-CAM
Engineering: Embedded Systems, MSP430, Arduino, ADPD4100, Biomedical Instrumentation, PCB Prototyping, Soldering
Software: Code Composer Studio, PyCharm, VS Code, Google Colab, GitHub, Fusion 360, Autodesk Inventor, COMSOL Multiphysics
Certifications: NASA Open Science 101 (NASA), R Programming, AutoDesk Inventor, OpenCV – PyTorch, | **Interests:** Rock Climbing

Scanned with CamScanner